Martin Pawelczyk

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Research interests

- Data-centric AI (Data curation for foundation model development, unlearning & privacy-preserving ML)
- Interpretability & AI Safety
- Evaluation & open source software

Positions

- '26 – **Assistant Professor for Responsible Computing, University of Vienna**.
- '23 – '25 **Postdoctoral Fellow, Harvard University**.
- '21 – '23 **Research Fellow, Harvard University**.
- '21 – '22 **Research Fellow, Max-Planck-Institute for Security & Privacy, Bochum**
- '21 **AI Research Intern, JP Morgan, London**

Education

- '18 – '23 **Ph.D. Computer Science, University of Tübingen, Germany**.
  On the Generation of Realistic and Robust Counterfactual Explanations for Algorithmic Recourse
  Advisor: Gjergji Kasneci, Reviewer: Ulrike von Luxburg.
- '17 – '18 **M.Sc. Statistics (Research), London School of Economics (LSE), UK**.
- '16 – '17 **M.Sc. Econometrics, University of Edinburgh, UK**.
- '13 – '16 **B.Sc. Economics, University of Cologne, Germany**.

Selected Honors and Awards

- '25 **Appointment as Assistant Professor**, UCL, NTU Singapore & TU Darmstadt (all declined)
- '24 **Spotlight Paper** 🏆, AAAI'24
- '21, '23, '24 **Spotlight Presentation** 🏆, ICML'21 WS, ICLR'22 WS, ICML'24 WS
- '22, '24 **Top Reviewer Award** 🏆, NeurIPS'22, NeurIPS'24
- '16-'18 **German National Merit Foundation Scholarship**, 40.000 Euros
- '16-'17 **University of Edinburgh Scholarship**, 10.000 Euros

Research Publications

H-index: **16**, Citations: **5028**

Conference Proceedings

1. Bordt, S., & **Pawelczyk**, M. (2026). Train Once, Answer All: Many Pretraining Experiments for the Cost of One. In *International Conference on Learning Representations* (***ICLR***).

2. **Pawelczyk**, M., Sun, L., Qi, Z., Kumar, A., & Lakkaraju, H. (2026). Generalizing Trust: Weak-to-Strong Trustworthiness in Language Models†. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics* (***ACL***)
   Spotlight Presentations 🏆 *at ICML'25 MOSS & CFAgentic Workshops*.

3. Karchmer, A., Neel, S., & **Pawelczyk**, M. (2025). Efficiently Verifiable Proofs of Data Attribution. In *Advances in Neural Information Processing Systems* (***NeurIPS***).

4. **Pawelczyk**, M., Di, J. Z., Lu, Y., Kamath, G., Sekhari, A., & Neel, S. (2025). Machine Unlearning Fails to Remove Data Poisoning Attacks†. In *International Conference on Learning Representations (**ICLR**),*
Spotlight Presentation 🏆 *at ICML'24 GenLaw Workshop*.

5. Leemann, T., **Pawelczyk**, M., Eberle, C. T., & Kasneci, G. (2024). I Prefer Not To Say: Protecting User Consent in Models with Optional Personal Data. In *38th Annual AAAI Conference on Artificial Intelligence (**AAAI**),*
Spotlight Presentation 🏆.

6. **Pawelczyk**, M., Neel, S., & Lakkaraju, H. (2024). In-Context Unlearning: Language Models as Few Shot Unlearners. In *International Conference on Machine Learning (**ICML**).*

7. Leemann*, T., **Pawelczyk***, M., & Kasneci, G. (2023). Gaussian Membership Inference Privacy. In *Advances in Neural Information Processing Systems (**NeurIPS**).*

8. **Pawelczyk**, M., Lakkaraju*, H., & Neel*, S. (2023). On the Privacy Risks of Algorihtmic Recourse. In *International Conference on Artificial Intelligence and Statistics (**AISTATS**).*

9. **Pawelczyk**, M., Leemann, T., Biega*, A., & Kasneci*, G. (2023). On the Trade-Off between Actionable Explanations and the Right to be Forgotten. In *International Conference on Learning Representations (**ICLR**).*

10. **Pawelczyk**, M., Datta, T., van-den-Heuvel, J., Kasneci, G., & Lakkaraju, H. (2023). Probabilistically Robust Recourse: Navigating the Tradeoffs between Costs and Robustness in Algorithmic Recourse†. In *International Conference on Learning Representations (**ICLR**).*

11. V. Borisov and K. Seßler and T. Leemann and M. **Pawelczyk** and G. Kasneci. (2023). Large Language Models are Realistic Tabular Data Generators†. In *International Conference on Learning Representations (**ICLR**).*

12. Agarwal, C., Saxena, E., Krishna, S., **Pawelczyk**, M., Johnson, N., Puri, I., … Lakkaraju, H. (2022). OpenXAI: Towards a Transparent Evaluation of Model Explanations. In *Advances in Neural Information Processing Systems (**NeurIPS**).*

13. Borisov, V., Leemann, T., Seßler, K., Haug, J., **Pawelczyk**, M., & Kasneci, G. (2022). Deep Neural Networks and Tabular Data: A Survey†. In *IEEE Transactions on Neural Networks and Learning Systems (**TNNLS**).*

14. **Pawelczyk**, M., Agarwal, C., Joshi, S., Upadhyay, S., & Lakkaraju, H. (2022). Exploring Counterfactual Explanations Through the Lens of Adversarial Examples: A Theoretical and Empirical Analysis. In *International Conference on Artificial Intelligence and Statistics (**AISTATS**).*

15. **Pawelczyk**, M., Bielawski, S., Van den Heuvel, J., Richter*, T., & Kasneci*, G. (2021). CARLA: A Python Library to Benchmark Algorithmic Recourse and Counterfactual Explanation Algorithms†. In *Advances in Neural Information Processing Systems (**NeurIPS**),*
Spotlight Presentation 🏆 *at ICML'21 Algorithmic Recourse Workshop*.

16. Haug, J., **Pawelczyk**, M., Broelemann, K., & Kasneci, G. (2020). Leveraging model inherent variable importance for stable online feature selection. In *Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining (**KDD**).*

17. **Pawelczyk**, M., Broelemann, K., & Kasneci, G. (2020). On Counterfactual Explanations under Predictive Multiplicity. In *36th Conference on Uncertainty in Artificial Intelligence (**UAI**).*

18. **Pawelczyk**, M., Broelemann, K., & Kasneci, G. (2020). Learning Model-Agnostic Counterfactual Explanations for Tabular Data. In *Proceedings of The Web Conference 2020 (**WWW**).*

Preprints

1. Abedini, S., Mavali, S., Schönherr, L., **Pawelczyk***, M., & Burkholz*, R. (2026). Don't Trust Stubborn Neighbors: A Security Framework for Agentic Networks. *arXiv preprint: arXiv:2603.15809.*

2. Andrii, S., Plant, C., & **Pawelczyk**, M. (2026). Forget What Comes Next: LLM Unlearning via Statistical Independence. *under review at ICLR 2027.*

3. Baniecki, H., Biecek, P., & **Pawelczyk**, M. (2026). LinMix: Data Mixing as Linear Domain Attribution. *under review at ICLR 2027.*

4. Bause, F., Niederle, J., **Pawelczyk**, M., & Burkholz, R. (2026). Multi-Agent Systems are Mixtures of Experts: Who Becomes an Influencer? *arXiv preprint: arXiv:2605.25929*.

5. König, G., **Pawelczyk**, M., Schönherr, L., von Luxburg, U., & Bordt, S. (2026). Validity Threats for Foundation Model Research. *arXiv preprint: arXiv:2606.05029*.

6. **Pawelczyk**, M., Rodriguez-Beltran, S., & Bordt, S. (2026). A Ground-Truth Model Suite for Unlearning Evaluation Across the LLM Training Lifecycle. *under review at ICLR 2027*.

Grants

- Co-Investigator of the grant: "Egocentric perception, interaction and computing in the deep learning era"
  German Academic Exchange Service (DAAD)
  Funding Amount: 80.000€
  2020 – 2022

Open Source Software Projects

'26 ◇ `TOAA`: Enabling academia to contribute to pretraining research
`https://github.com/sbordt/pretrain-experiments/`

'23 ◇ `GReaT`: Generation of Realistic Tabular data with LLMs to enable private data sharing
`https://github.com/kathrinse/be_great`, **350+ github stars**

'22 ◇ `OpenXAI`: A library to evaluate feature attribution algorithms
`https://github.com/AI4LIFE-GROUP/OpenXAI`, **250+ github stars**

'21 ◇ `CARLA`: A library to evaluate counterfactual explanation algorithms
`https://github.com/carla-recourse/CARLA`, **300+ github stars**

Presentations

'26 ◇ *Beyond Retraining: Efficient Algorithms and Evaluations for Private Data Unlearning*
University of Virginia, Lecture, Remote
Huawei Aurora Tech Summit, Keynote Talk, Helsinki

'25 ◇ *Verifying Data Attributions without Breaking the Bank*,
Winter School on Causality and Explainable AI, Sorbonne University, Paris

◇ *Privacy-Utility Tradeoffs in LLM Systems: From Data Selection to Agent Behavior*,
University College London (UCL), London

◇ *Building AI Safeguards: Computational Methods for Privacy & Explainability Compliance*,
ETH Zurich, Zurich
University of Vienna, Vienna
TU Darmstadt, Darmstadt
Technical University of Munich (TUM), Heilbronn
Nanyang Technological University (NTU), Singapore
CISPA Helmholtz Center for Information Security, Saarbrücken
STOR Colloquium, University of North Carolina (UNC), Chapel Hill
Responsible Data Science Lab, hosted by Prof. Gjergji Kasneci, TU Munich (TUM)

'24 ◇ *Machine Unlearning Fails to Remove Data Poisoning Attacks*,
Spotlight Presentation 🏆, GenLaw Workshop, ICML 2024, Vienna

◇ *Unlearning Private Data from Large Language Models: Algorithms and Evaluations*,
DSE Lab, hosted by Prof. Jiliang Tang, Michigan State University

◇ *Guarding Private Data from Membership Inference Adversaries*,
Privacy Workshop, TU Delft, Netherlands

Presentations (continued)

- *On the Privacy Risks of Counterfactual Explanations for Algorithmic Recourse*,
  Workshop on Interpretability and Algorithmic Recourse, Saarland University, Germany

'23
- *On the Tradeoffs betwenn Actionable Explantions and the Right to be Forgotten*,
  Max-Plank Institute for Privacy & Security, Bochum, Germany
  LMU Munich, Statistical Data Science Seminar
- *On the Privacy Risks of Counterfactual Explanations for Algorithmic Recourse*,
  TUM Reliable Data Science Seminar, Technical University of Munich, Germany
  2. XAI Winter School, Nice, France

'22
- *Lecture on Realistic & Robust Counterfactual Explanations for Algorithmic Recourse*,
  XAI Summer School, TU Delft, Netherlands
- *On the Tradeoffs between Actionable Explantions and the Right to be Forgotten*,
  1. XAI Winter School, Nice, France

'21
- *CARLA: A Python Library to Benchmark Counterfactual Explanation Algorithms*,
  Spotlight Presentation 🏆, Algorithmic Recourse Workshop, ICML 2021
- *On Realistic & Robust Counterfactual Explanations for Algorithmic Recourse*,
  Harvard University, AI4Life Lab, Remote
  JP Morgan AI Research, London, UK

Teaching

WS '18/'19 ◇ **Data Mining and Probabilistic Reasoning**, **Teaching assistant**, University of Tübingen

SS '19 ◇ **Trustworthy Machine Learning, Co-Lecturer**, University of Tübingen

WS '19/'20 ◇ **Data Mining and Probabilistic Reasoning, Teaching assistant**, University of Tübingen

SS '20 ◇ **Trustworthy Machine Learning, Co-Lecturer**, University of Tübingen

SS '26 ◇ **Mining Massive Data, Lecturer**, University of Vienna

WS '26/'27 ◇ **Building Foundation Models from Scratch, Lecturer**, University of Vienna

Research Mentoring

'26 ◇ Andrii Shkabrii, Aleksei Fokin, Sebastian Rodriguez Beltran (all University of Vienna),
Hubert Baniecki (University of Warsaw)

'25 ◇ Samira Abedini (Saarland University), Sina Mavali (CISPA)

'24 ◇ Lillian Sun (ACL'26), Catherine Huang (BSc students, all Harvard University)

'23 ◇ Jimmy Z. Di (MSc student, University of Waterloo; ICLR'25)

'22 ◇ Vadim Borisov, Kathrin Sessler, Tobias Leemann (all University of Tübingen; ICLR'23, NeurIPS'23, AAAI'24)

'21 ◇ Teressa Datta (MSc student, Harvard University; ICLR'23)

◇ Johan van-den Heuvel (MSc student, University of Tübingen; ICLR'23 & NeurIPS'21)

'20 ◇ Sascha Bielawski (MSc student, University of Tübingen; NeurIPS'21)

Workshops and Tutorials

2025 ◇ **ICLR workshop**
"Building Trust in LLMs and LLM Applications: From Guardrails to Explainability to Regulation"
https://building-trust-in-llms.github.io/iclr-workshop/
with Hima Lakkaraju, Micah Goldblum, Ramasuri Narayanam, Soumyabrata Pal and Bang An

Workshops and Tutorials (continued)

2022 ◇ **Tutorial at XAI Summer School**
CARLA: A Python Library to Benchmark Counterfactual Explanation Algorithms
TU Delft, Netherlands

Reviewing and Service

Area Chair
- ◇ 2027: ICML
- ◇ 2026: ICLR, KDD, NeurIPS

Reviewer
- ◇ 2026: SatML
- ◇ 2025: ICLR, ICML, NeurIPS, TMLR
- ◇ 2024: ICLR, ICML, NeurIPS (outstanding reviewer 🏆), TMLR
- ◇ 2023: AISTATS, FAccT, ICML, JMLR, NeurIPS & NeurIPS DSB Track, ICLR
- ◇ 2022: AISTATS, AAAI, NeurIPS (outstanding reviewer 🏆) & NeurIPS DSB Track, ICML, FAccT
- ◇ 2021: NeurIPS & NeurIPS DSB Track, AAAI

References

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Prof. Seth Neel, PhD
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